ECE 487
HOMEWORK ~~8~~ 7
SOLUTIONS

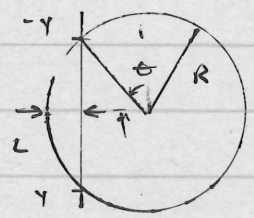
1 a) OVERSPREAD IF $f_0 < \frac{c^2}{8R^2\Omega}$ $\qquad \Omega = \frac{2\pi}{1.03 \text{ days}}$

$= 7.06 \times 10^{-5} \text{ s}^{-1}$

$\overset{?}{<} 137.8 \text{ MHz}$ $\qquad R = 3400 \text{ km}$

NEITHER S-BAND NOR UHF RADAR CAN BE USED TO MAP ENTIRE SURFACE OF MARS WHICH IS OVERSPREAD FOR BOTH

b) TAKE $f_0 = 2.38 \text{ GHz}$



SELECT IPP S.T. $\delta t = \frac{2L}{c}$ $\qquad L = R(1-\cos\theta)$

ALSO, $\frac{1}{2\delta t} = \frac{2\Omega y f_0}{c}$ $\qquad y = R\sin\theta$

EQUATE THESE TWO: $\frac{c}{4L} = \frac{2y\Omega f_0}{c}$

$\frac{1}{R(1-\cos\theta)} = \frac{8\Omega f_0}{c^2} R\sin\theta$

SO $(1-\cos\theta)\sin\theta = \frac{c^2}{8R^2\Omega f_0}$

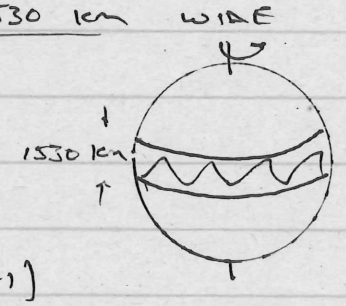
$\frac{\theta^3}{2} = 0.0058$ $\qquad$ SAY θ SMALL, $\cos\theta \sim 1 - \frac{\theta^2}{2}$

$\sin\theta \sim \theta$

$\theta = 0.226 \text{ RAD} \sim 13°$

SO $L = 87 \text{ km}$, $y = 765 \text{ km}$

FINALLY, CAN MAP A STRIP OF MARS $2y \cong$ <u>1530 km</u> WIDE



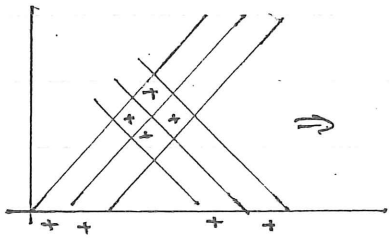
c) MAX DOPPLER SHIFT IS $f_D = \frac{2y_{max}\Omega f_0}{c}$

$= 857 \text{ Hz}$

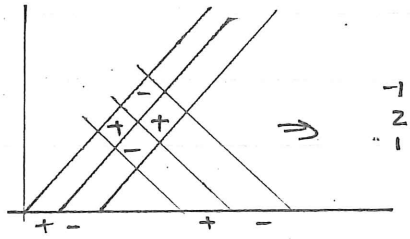
SO $\text{IPP} = \frac{1}{2 \cdot 857} = \underline{0.583 \text{ ms}} = (4 \text{ ms})(\underbrace{2^p - 1}_{\sim 145})$

TAKE $p = 7$ AND GET REASONABLY CLOSE

3.



⇒ 1 2 1



⇒ −1 2 1

SUMMING ⇒ 0 4 0

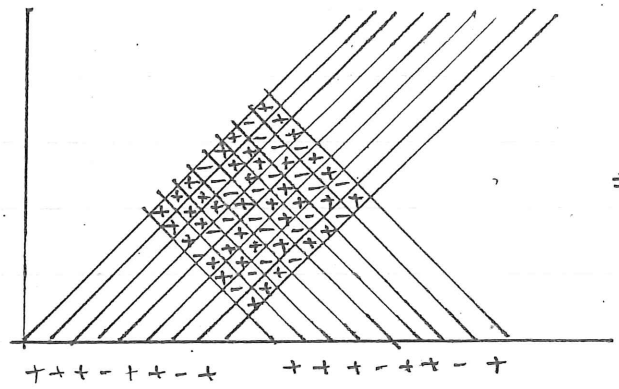
MAIN LOBE = 4,

NO RANGE SIDELOBES

LONGER CODES: 4-BIT + + + −

+ + − +

3-CONT

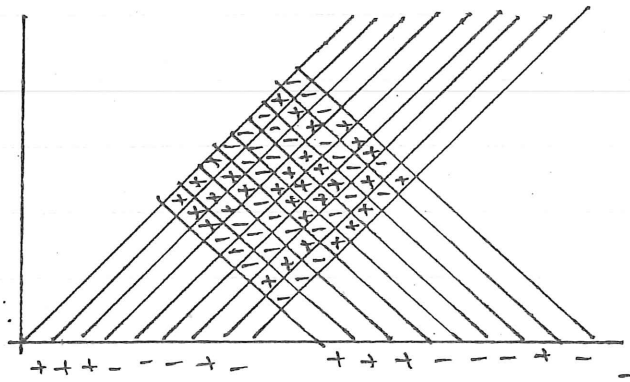


⇒ 1 0 1 0 3 0 −1 8 −1 0 3 0 1 0 1

← MAIN LOBE

+ RANGE SIDELOBES

NEXT IPP



−1 0 −1 0 −3 0 1 8 1 0 −3 0 −1 0 −1

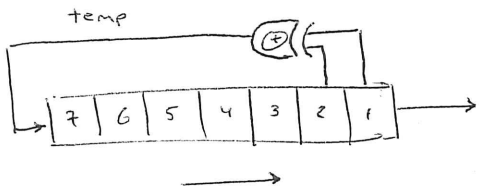
← MAIN LOBE

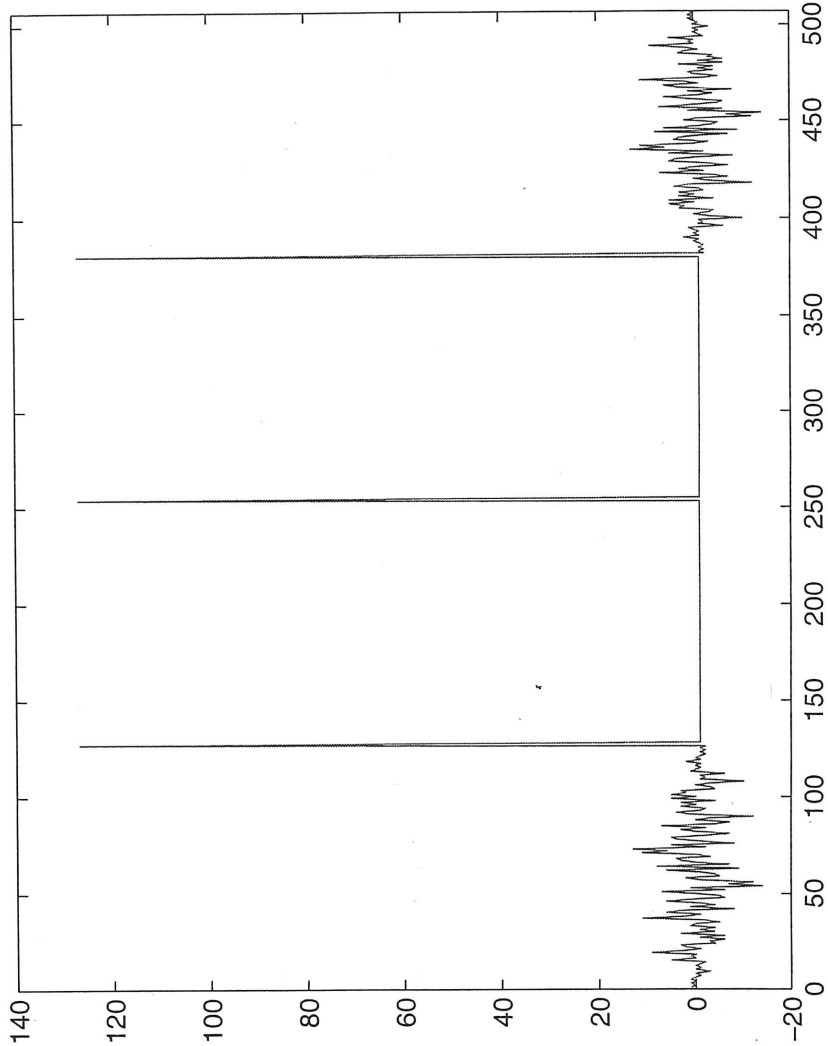
+ RANGE SIDELOBES

SUM

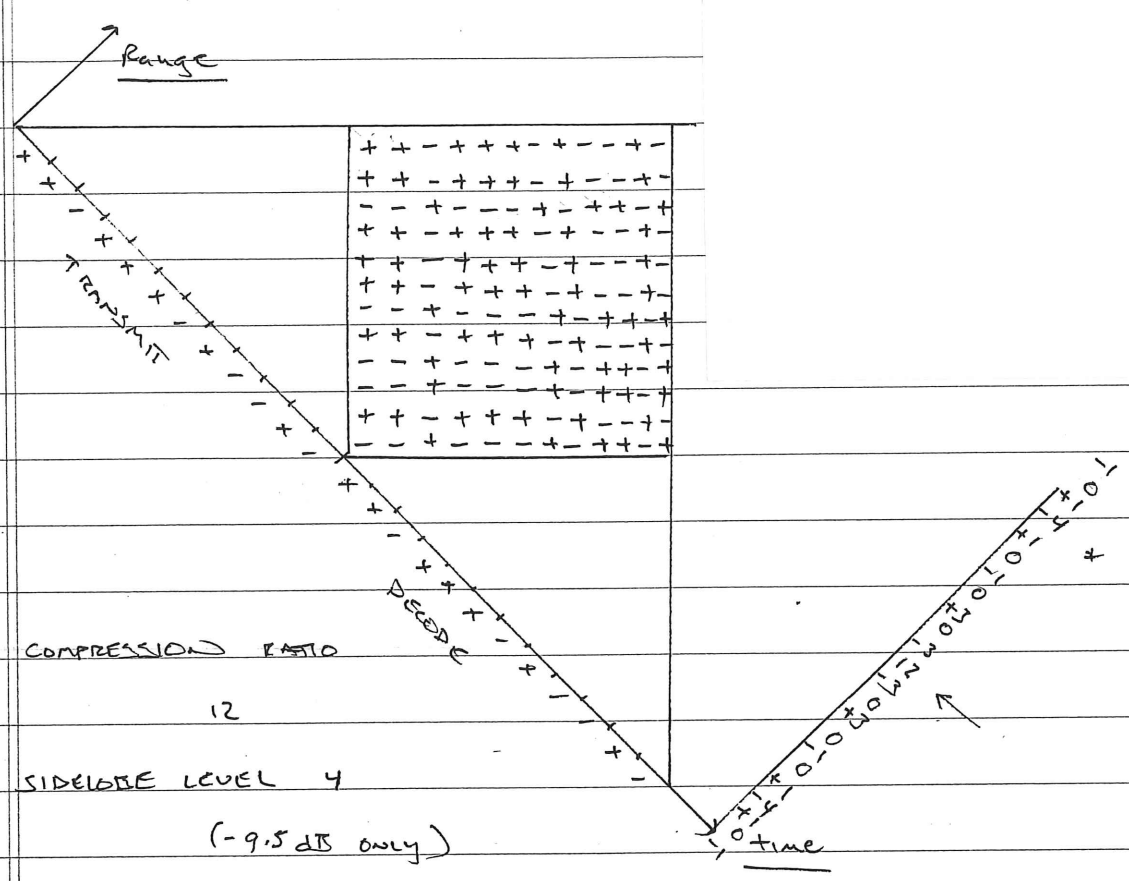
MAIN LOBES ADD TO 16, RANGE SIDELOBES <u>VANISH</u>

```
% problem 2: cyclic codes

nb=7;
nc=2^7-1; % sequence repeats here
bits = ones(nb,1);
code1 = zeros(nc,1);  % time reverse of single code sequence
code2 = zeros(nc*3,1);  % repeating code sequence

for i=1:nc

    temp=xor(bits(1),bits(2)); % set

    for j=2:nb
        bits(j-1)=bits(j);  % shift
    end

    bits(nb)=temp;

    code1(nc+1-i)=bits(1);
    code2(i)=bits(1);
end

code1=2*(code1-0.5);
code2=2*(code2-0.5);

code2(nc+1:2*nc)=code2(1:nc);
code2(2*nc+1:3*nc)=code2(1:nc);

cf=conv(code1,code2);  % compute correlation using convolution

plot(cf);
xlim([0 507]);
```





2 B_y = ++-+ B_3 = ++- B_{y3} = ++-+ ++-+ --+-



COMPRESSION RATIO

12

SIDELOBE LEVEL 4

(-9.5 dB only)